SpendWise

A Python-based Personal Expense Analyzer

Group Project Report

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1. Introduction

As students we spend money on many small things like food orders, auto rides and recharges, and by the end of the month it is hard to say where it all went. SpendWise is a small Python program that reads expenses from a CSV file, puts each transaction into a category, stores everything in an SQLite database and shows the spending with charts. It also lets us set a budget for each category and tells us in which months we went over it.

The project uses Pandas for handling the data, Matplotlib for the charts and the built-in sqlite3 module for storage. Categorization is done with simple keyword rules. It is our own implementation and is not based on any existing project.

2. Objectives

• Import expense records from a CSV file.

• Fill missing categories automatically using keyword rules.

• Find the total spend and category-wise spend.

• Draw monthly and category charts.

• Store transactions in SQLite and query them with SQL.

• Set monthly budgets per category and detect overspending.

• Compute basic statistics and flag unusual expenses using IQR.

• Produce a summary report as text and CSV.

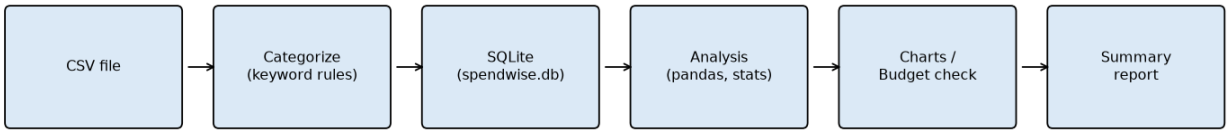
3. Dataset

We did not use any external dataset. A script generate_data.py creates synthetic transactions for January to June 2026 with a fixed random seed (42), so the same file is produced every time. It has 294 rows in data/expenses.csv. Descriptions are things a hostel student would actually pay for, like "Swiggy order - biryani", "Ola auto ride", "Mobile recharge Jio" and "Room rent". Rent is Rs 6000 every month. About 40% of rows (121 rows) have the category left blank on purpose so the categorizer has something to do.

Column	Type	Example
date	YYYY-MM-DD	2026-01-07
description	text	Myntra - shoes
category	text / blank	Shopping
amount	integer (INR)	2116

4. System Design

The program works as a simple pipeline. The CSV is loaded, blank categories are filled, the clean data is saved to SQLite, and then the analysis, charts, budget check and report are produced from it.



Helper functions are kept in spendwise.py and the notebook SpendWise.ipynb calls them step by step.

5. Implementation

5.1 Importing the CSV

The file is read with pandas, dates are parsed, and a month column is added for grouping.

```
df = pd.read_csv(path, parse_dates=['date'])
df['category'] = df['category'].fillna('').astype(str).str.strip()
df['month'] = df['date'].dt.to_period('M').astype(str)
```

5.2 Auto-categorization

A dictionary maps each category to keywords. The description is lowercased and checked against each list. If nothing matches, the row goes to Others.

```
RULES = {'Food': ['swiggy', 'zomato', 'canteen', ...],
         'Transport': ['uber', 'ola', 'petrol', 'bus', 'train'], ...}

def categorize(desc):
    text = desc.lower()
    for cat, words in RULES.items():
        if any(w in text for w in words):
            return cat
    return 'Others'
```

5.3 Totals

```
total = df['amount'].sum()
cat_total = df.groupby('category')['amount'].sum()
```

5.4 Charts

Three charts are made: monthly bar chart, category pie chart and a stacked bar using a pivot table.

```
pivot = df.pivot_table(index='month', columns='category',
                       values='amount', aggfunc='sum', fill_value=0)
pivot.plot(kind='bar', stacked=True)
```

5.5 SQLite storage

```
con = sqlite3.connect('spendwise.db')
df.to_sql('transactions', con, if_exists='replace', index=False)
pd.read_sql_query('SELECT category, SUM(amount) FROM transactions
                   GROUP BY category', con)
```

5.6 Budget check

Spending is grouped by month and category and compared with the budget dictionary.

```
spent = df.groupby(['month', 'category'])['amount'].sum().reset_index()
spent['budget'] = spent['category'].map(budgets)
over = spent[spent['amount'] > spent['budget']]
```

5.7 Statistics and report

Mean, median and standard deviation come from pandas. An expense is unusual if it is above Q3 + 1.5 x IQR. Rent is excluded since it is a fixed payment. The summary is written to summary_report.txt, summary.csv and overspending.csv.

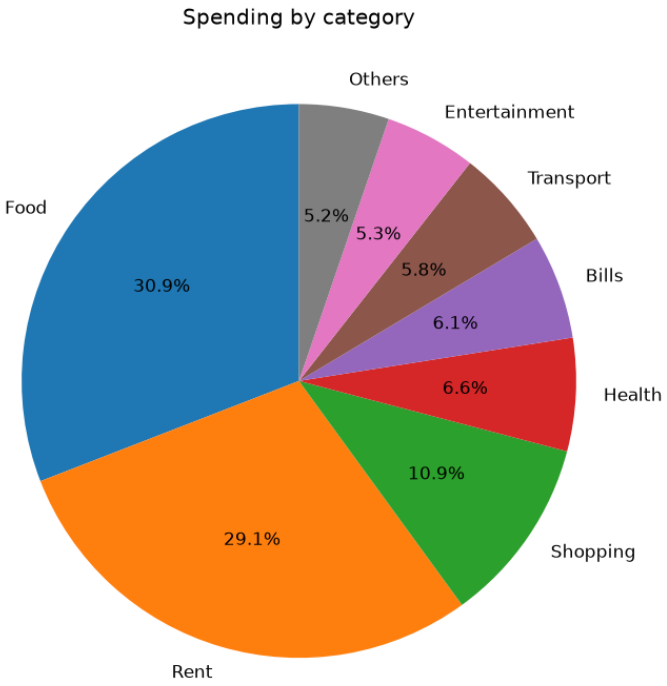
```
q1, q3 = df['amount'].quantile([0.25, 0.75])
limit = q3 + 1.5 * (q3 - q1)
```

6. Results

Total spend over six months was **Rs 123,656** across 294 transactions, which is about Rs 20,609 per month. The rules filled 121 blank categories.

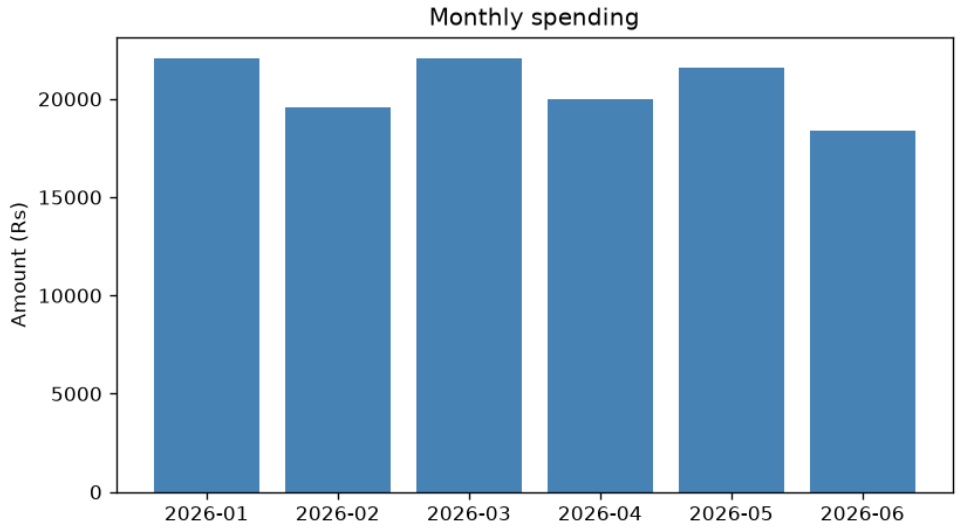
Category	Total (Rs)	Share
Food	38,201	30.9%
Rent	36,000	29.1%
Shopping	13,480	10.9%
Health	8,161	6.6%
Bills	7,517	6.1%
Transport	7,210	5.8%
Entertainment	6,596	5.3%
Others	6,491	5.2%

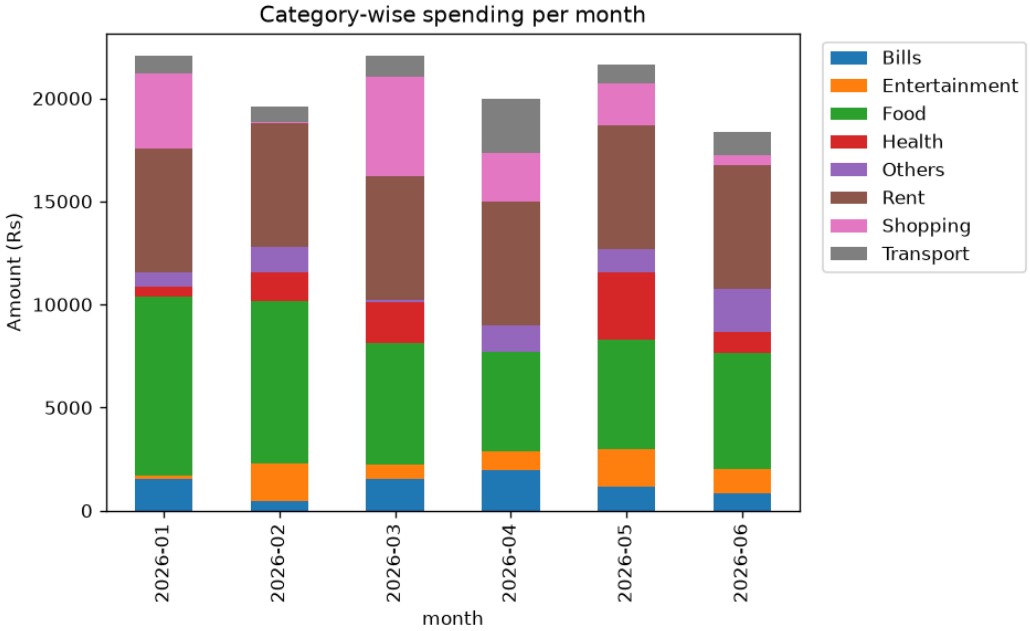
Food was the biggest category at Rs 38,201, slightly more than rent (Rs 36,000). Shopping was third at Rs 13,480.



Month	2026-01	2026-02	2026-03	2026-04	2026-05	2026-06
Spend (Rs)	22,057	19,599	22,041	19,969	21,618	18,372

January was the highest month (Rs 22,057) and June the lowest (Rs 18,372).





6.1 Budget overspending

Budgets used (Rs per month): Food 7000, Transport 2000, Shopping 3000, Bills 1500, Entertainment 1500, Health 1500, Rent 6000, Others 1500.

Month	Category	Spent	Budget	Over by
2026-01	Bills	1,529	1,500	29
2026-01	Food	8,705	7,000	1,705
2026-01	Shopping	3,646	3,000	646
2026-02	Entertainment	1,810	1,500	310
2026-02	Food	7,890	7,000	890
2026-03	Bills	1,554	1,500	54
2026-03	Health	1,975	1,500	475
2026-03	Shopping	4,814	3,000	1,814
2026-04	Bills	1,944	1,500	444

2026-04	Transport	2,614	2,000	614
2026-05	Entertainment	1,851	1,500	351
2026-05	Health	3,300	1,500	1,800
2026-06	Others	2,120	1,500	620

There were 13 cases of overspending. Every month went over in at least one category. The largest were Health in May 2026 (Rs 1,800 over), Shopping in March 2026 (Rs 1,814 over) and Food in January 2026 (Rs 1,705 over). Bills went over in Jan, Mar and Apr, but only slightly in two of them.

6.2 Statistics

Measure	Value (Rs)
Mean	420.60
Median	225.5
Std deviation	863.00
Q1 / Q3	101.5 / 435.0
IQR upper limit	935.25

The top 5 expenses overall are all the monthly rent of Rs 6000. Leaving rent out, the top 5 are:

Date	Description	Category	Amount
2026-01-07	Myntra - shoes	Shopping	2,116
2026-05-21	Myntra - shoes	Shopping	1,908
2026-03-07	Myntra - shoes	Shopping	1,812
2026-06-12	Zomato - pizza	Food	1,764
2026-04-14	Electricity share	Bills	1,512

11 non-rent transactions were above the IQR limit of Rs 935.25 and were flagged as unusual, mostly shoe purchases, large food orders and a hospital visit. The mean (Rs 420.60) is much higher than the median (Rs 225.5) because of rent and these few large payments.

7. Conclusion and Future Work

SpendWise does what we planned: it imports a CSV, fills missing categories, stores the data in SQLite, makes charts, checks budgets and writes a summary. From our sample data we can see that food delivery is the main area to cut down, and that health and shopping costs come as occasional big spikes.

In future we would like to add a simple GUI or web page, let the user edit the keyword rules, read bank SMS or UPI statements directly, and try a machine learning classifier instead of fixed keywords. Predicting next month spending would also be useful.

8. References

pandas documentation, https://pandas.pydata.org/docs/

Matplotlib documentation, https://matplotlib.org/stable/

Python sqlite3 module, https://docs.python.org/3/library/sqlite3.html

SQLite documentation, https://www.sqlite.org/docs.html